

Nmap — Practical Tool Guide

From host and port discovery to service/version detection and NSE scripts, with the key options explained and when to use them.

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Usage note: Focused on practical review commands for eJPT labs. Targets in examples are placeholders, not real targets.

DISCOVER

Know the target

PORTS

Find open ports

SERVICE

Identify service/version

NSE

Run a suitable script

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1. Getting started and understanding the syntax

Start from the question you need to answer, then choose the scan type instead of memorizing one command for every situation.

KALI TERMINAL — Quick help

```
nmap -h
```

This shows the main options. Use `man nmap` for the complete manual.

Option	Meaning	When to use it
-sn	Host discovery only; no port scan	Find live hosts in an authorized lab subnet.
-Pn	Skip host discovery and assume the target is up	When ICMP/ping is blocked or the target does not answer ping.
-p	Specify a port, list, or range	For example -p22,80,443 or -p1-1000.
-p-	Scan all TCP ports 1-65535	Avoid missing a service on a non-standard port.
--top-ports N	Scan the N most common ports	Faster first-pass scan.

2. Important scan types

These are the options worth recognizing and why they are used.

Option	Meaning	When to use it
-sS	TCP SYN scan	Common and fast TCP scan; requires suitable privileges.
-sT	TCP connect scan	Alternative when raw SYN packets cannot be used.
-sU	UDP scan	For UDP services such as DNS or SNMP; usually slower.
-sV	Service/version detection	After finding open ports, identify the service and version where possible.
-O	OS detection	Estimate the operating system from network fingerprints; the result is an estimate.
-sC	Default NSE scripts	Run the default script set; review scope because some scripts can be intrusive.
-A	OS + version + scripts + traceroute	Useful in a lab when you want more information quickly; noisier.
-T3 / - T4	Timing template	T4 is common in a fast, stable lab. Do not raise timing without a reason on real networks.

3. Practical scan workflow

Use two stages: discover the ports, then focus on the services you found.

KALI TERMINAL — Common ports first

```
nmap -Pn --top-ports 1000 [TARGET]
```

KALI TERMINAL — Full TCP scan inside a lab

```
nmap -Pn -p- -T4 [TARGET]
```

Note: A full scan can produce substantial traffic. Use it only inside the authorized scope.

KALI TERMINAL — Re-scan found ports with versions and default scripts

```
nmap -Pn -sV -sC -p22,80,445 [TARGET]
```

KALI TERMINAL — Targeted UDP instead of scanning every UDP port

```
sudo nmap -Pn -sU -p53,69,111,123,161 [TARGET]
```

Prioritize expected UDP services first; a full UDP scan is slow.

4. NSE scripts: how to find and run them

Instead of memorizing hundreds of scripts, connect the service name to the script family and read its help first.

KALI TERMINAL — Script directory on Kali

```
ls /usr/share/nmap/scripts | head
```

KALI TERMINAL — Search by service/name

```
ls /usr/share/nmap/scripts | grep -i "^smb-"  
ls /usr/share/nmap/scripts | grep -i "http"
```

KALI TERMINAL — Read script help before running it

```
nmap --script-help smb-os-discovery  
nmap --script-help "http-*
```

KALI TERMINAL — Run one script on one port

```
nmap -Pn -p445 --script smb-os-discovery [TARGET]
```

KALI TERMINAL — Run a service family

```
nmap -Pn -p80,443 --script "http-*" [TARGET]
```

Note: A wildcard is broad. Review `nmap --script-help "http-"` first because the family can contain scripts that are not passive.

KALI TERMINAL — Pass script arguments

```
nmap -Pn -p22 --script ssh-auth-methods --script-args="ssh.user=[USER]" [TARGET]
```

KALI TERMINAL — Refresh the NSE database after adding a local script

```
sudo nmap --script-updatedb
```

5. How to search correctly “by port”

NSE execution is normally connected to the service/portrule. The port points you to the service name; then you search for scripts for that service.

KALI TERMINAL — Example: port 445 is open

```
nmap -Pn -sV -p445 [TARGET]
ls /usr/share/nmap/scripts | grep -i "^smb-"
nmap --script-help "smb-"
```

KALI TERMINAL — Run appropriate SMB enumeration scripts on 445

```
nmap -Pn -p445 --script "smb-enum-*" [TARGET]
```

Note: Do not search only for a “script for port 445.” First identify SMB, then search smb-*. The same idea applies to http-*, ftp-*, ssh-*, and other service families.

6. Output and saving results

Save scan results because SearchSploit and Metasploit can use service/version data later.

KALI TERMINAL — Normal text

```
nmap -Pn -sV -p[PORTS] -oN nmap.txt [TARGET]
```

KALI TERMINAL — All main formats

```
nmap -Pn -sV -p[PORTS] -oA scan [TARGET]
```

This creates .nmap, .gnmap, and .xml outputs.

KALI TERMINAL — XML for SearchSploit

```
nmap -Pn -sV -oX nmap.xml [TARGET]
searchsploit --nmap nmap.xml
```

7. Common mistakes

- Understanding the result matters more than using many flags.
- Open ports do not mean OS detection. -O is OS detection, -sV is service/version detection, and -sU is UDP scanning.
- An open port is not proof of a vulnerability. Use the service/version as evidence for enumeration and CVE/exploit research.
- Do not run --script all or brute/dos/exploit/intrusive script categories randomly. NSE is not a sandbox and can affect the target.

